13. Variable Stars

Stars with changing magnitudes are called *variables* (Fig. 13.1). Variations in the brightness of stars were first noted in Europe at the end of the 16th century, when *Tycho Brahe's supernova* lit up (1572) and the regular light variation of the star o Ceti (Mira) was observed (1596). The number of known variables has grown steadily as observational precision has improved (Fig. 13.2). The most recent catalogues contain about 40,000 stars known or suspected to be variable.

Strictly speaking, all stars are variable. As was seen in Chap. 11, the structure and brightness of a star change as it evolves. Although these changes are usually slow, some evolutionary phases may be extremely rapid. In certain evolutionary stages, there will also be periodic variations, for example pulsations of the outer layers of a star.

Small variations in stellar brightness are also caused by hot and cool spots on a star's surface, appearing and disappearing as it rotates about its axis. The luminosity of the Sun changes slightly because of the sunspots. Probably there are similar spots on almost all stars.

Initially stellar brightnesses were determined visually by comparing stars near each other. Later on, comparisons were made on photographic plates. At present the most accurate observations are made photoelectrically or using a CCD camera. The magnitude variation as a function of time is called the *lightcurve* of a star (Fig. 13.3). From it one obtains the *amplitude* of the magnitude variation and its *period*, if the variation is periodic.

The basic reference catalogue of variable stars is the *General Catalogue of Variable Stars* by the Soviet astronomer *Boris Vasilyevich Kukarkin*. New, supplemented editions appear at times; the fourth edition published in 1985–1987, edited by P.N. Kholopov, contains about 32,000 variables of the Milky Way galaxy.

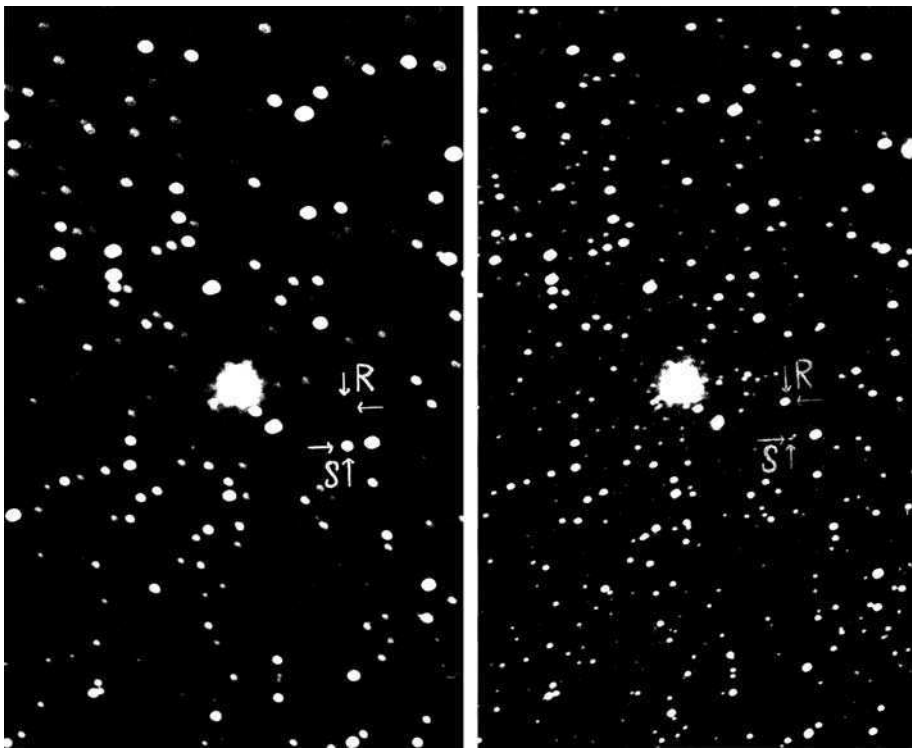

Fig. 13.1. The variables are stars changing in brightness. Two variables in Scorpius, R and S Sco (Photograph Yerkes Observatory)

Hannu Karttunen et al. (Eds.), *Variable Stars*.
In: Hannu Karttunen et al. (Eds.), Fundamental Astronomy, 5th Edition. pp. 279–290 (2007)
DOI: 11685739_13 © Springer-Verlag Berlin Heidelberg 2007

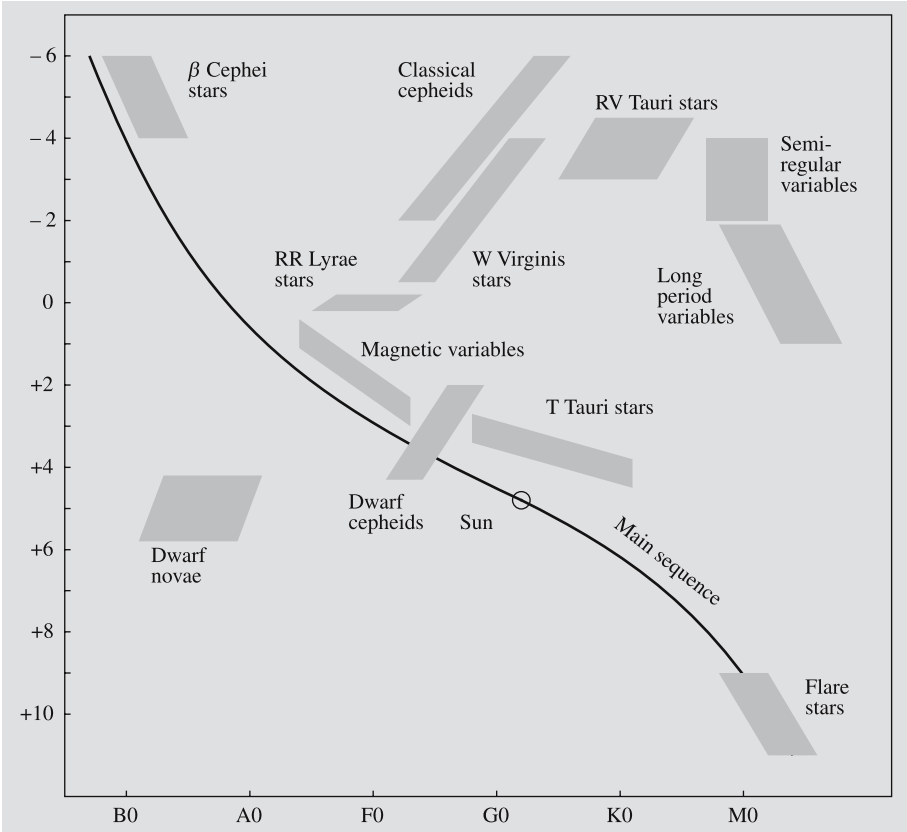

Fig. 13.2. The location of variables in the HR diagram

13.1 Classification

When a new variable is discovered, it is given a name according to the constellation in which it is located. The name of the first variable in a given constellation is R, followed by the name of the constellation (in the genitive case). The symbol for the second variable is S, and so on, to Z. After these, the two-letter symbols RR, RS, . . . to ZZ are used, and then AA to QZ (omitting I). This is only enough for 334 variables, a number that has long been exceeded in most constellations. The numbering therefore continues: V335, V336, etc. (V stands for variable). For some stars the established Greek letter symbol has been kept, although they have later been found to be variable (e. g. δ Cephei).

The classification of variables is based on the shape of the lightcurve, and on the spectral class and observed radial motions. The spectrum may also contain dark absorption lines from material around the star. Observations can be made outside the optical region as well. Thus the radio emission of some variables (e. g. flare stars) increases strongly, simultaneously with their optical brightness. Examples of radio and X-ray variables are the radio and X-ray pulsars, and the X-ray bursters.

Variables are usually divided into three main types: *pulsating*, *eruptive* and *eclipsing variables*. The eclipsing variables are binary systems in which the components periodically pass in front of each other. In these variables the light variations do not correspond to any physical change in the stars. They have been treated in connection with the binary stars. In the other variables the brightness variations are intrinsic to the stars. In the pulsating variables the variations are due to the expansion and contraction of the outer layers. These variables are giants and supergiants that have reached an unstable stage in their evolution. The eruptive variables are usually faint stars ejecting mass.

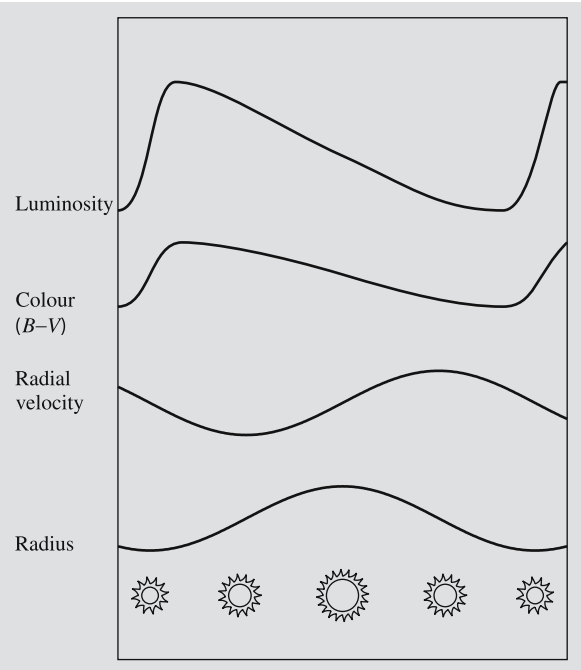

Fig. 13.3. The variation of brightness, colour and size of a cepheid during its pulsation

They are mostly members of close binary systems in which mass is transferred from one component to the other.

In addition a few *rotating variables* are known, where the brightness variations are due to an uneven temperature distribution on the surface, starspots coming into sight when the star rotates. Such stars may be quite common – after all, our Sun is a weak rotating variable. The most prominent group of rotating variables are the magnetic A stars (e. g. the α^2 Canum Venaticorum stars). These stars have strong magnetic fields that may be giving rise to starspots. The periods of rotating variables range from about 1 day to 25 d, and the amplitudes are less than 0.1 mag.

13.2 Pulsating Variables

The wavelengths of the spectral lines of the pulsating variables change along with the brightness variations (Table 13.1). These changes are due to the Doppler effect, showing that the outer layers of the star are indeed

Table 13.1. The main properties of pulsating variables (N, number of stars of the given type in Kukarkin's catalogue, P, pulsation period in days, Δm, pulsation amplitude in magnitudes)

Variable	N	P	Spectrum	Δm
Classical cepheids (δ Cep, W Vir)	800	1–135	F–K I	$\lesssim 2$
RR Lyrae	6100	< 1	A–F8	$\lesssim 2$
Dwarf cepheids (δ Scuti)	200	0.05–7	A–F	$\lesssim 1$
β Cephei	90	0.1–0.6	B1–B3 III	$\gtrsim 0.3$
Mira variables	5800	80–1000	M–C	$\gtrsim 2.5$
RV Tauri	120	30–150	G–M	$\lesssim 4$
Semiregular	3400	30–1000	K–C	$\lesssim 4.5$
Irregular	2300	–	K–M	$\lesssim 2$

pulsating. The observed gas velocities are in the range of 40–200 km/s.

The period of pulsation corresponds to a *proper frequency* of the star. Just like a tuning fork vibrates with a characteristic frequency when hit, a star has a fundamental frequency of vibration. In addition to the fundamental frequency other frequencies, "overtones", are possible. The observed brightness variation can be understood as a superposition of all these modes of vibration. Around 1920, the English astrophysicist *Sir Arthur Eddington* showed that the period of pulsation P is inversely proportional to the square root of the mean density,

$$P \propto \frac{1}{\sqrt{\rho}} \,. \tag{13.1}$$

The diameter of the star may double during the pulsation, but usually the changes in size are minor. The main cause of the light variation is the periodic variation of the surface temperature. We have seen in Sect. 5.6 that the luminosity of a star depends sensitively on its effective temperature, $L \propto T_e^4$. Thus a small change in effective temperature leads to a large brightness variation.

Normally a star is in stable hydrostatic equilibrium. If its outer layers expand, the density and temperature decrease. The pressure then becomes smaller and the force of gravity compresses the gas again. However, unless energy can be transferred to the gas motions, these oscillations will be damped.

The flux of radiative energy from the stellar interior could provide a source of energy for the stellar oscil-

lations, if it were preferentially absorbed in regions of higher gas density. Usually this is not the case but in the *ionization zones*, where hydrogen and helium are partially ionized, the opacity in fact becomes larger when the gas is compressed. If the ionization zones are at a suitable depth in the atmosphere, the energy absorbed during compression and released during expansion of an ionization zone can drive an oscillation. Stars with surface temperatures of 6000–9000 K are liable to this instability. The corresponding section of the HR diagram is called the cepheid instability strip.

Cepheids. Among the most important pulsating variables are the cepheids, named after δ Cephei (Fig. 13.3). They are population I supergiants (stellar populations are discussed in Sect. 17.2) of spectral class F–K. Their periods are 1–50 days and their amplitudes, 0.1–2.5 magnitudes. The shape of the light curve is regular, showing a fairly rapid brightening, followed by a slower fall off. There is a relationship between the period of a cepheid and its absolute magnitude (i. e. luminosity), discovered in 1912 by *Henrietta Leavitt* from cepheids in the Small Magellanic Cloud. This *period–luminosity relation* (Fig. 13.4) can be used to measure distances of stars and nearby galaxies.

We have already noted that the pulsation period is related to the mean density. On the other hand the size of a star, and hence its mean density, is related to its total luminosity. Thus one can understand why there should be a relation between the period and the luminosity of a pulsating star.

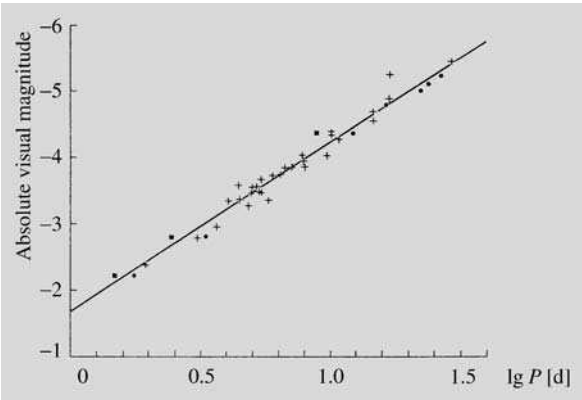

Fig. 13.4. The period–luminosity relation for cepheids. The black points and squares are theoretically calculated values, the crosses and the straight line represent the observed relation. (Drawing from Novotny, E. (1973): *Introduction to Stellar Atmospheres and Interiors* (Oxford University Press, New York) p. 359)

The magnitudes M and periods P of classical cepheids are shown in Fig. 13.4. The relation between M and $\log P$ is linear. However, to some extent, the cepheid luminosities also depend on colour: bluer stars are brighter. For accurate distance determinations, this effect needs to be taken into consideration.

W Virginis Stars. In 1952 *Walter Baade* noted that there are in fact two types of cepheids: the classical cepheids and the W Virginis stars. Both types obey a period–luminosity relation, but the W Vir stars of a given period are 1.5 magnitudes fainter than the corresponding classical cepheids. This difference is due to the fact that the classical cepheids are young population I objects, whereas the W Vir stars are old stars of population II. Otherwise, the two classes of variables are similar.

Earlier, the W Vir period–luminosity relation had been used for both types of cepheids. Consequently the calculated distances to classical cepheids were too small. For example, the distance to the Andromeda Galaxy had been based on classical cepheids, since only these were bright enough to be visible at that distance. When the correct period–luminosity relation was used, all extragalactic distances had to be doubled. Distances within the Milky Way did not have to be changed, since their measurements were based on other methods.

RR Lyrae Stars. The third important class of pulsating variables are the *RR Lyrae stars*. Their brightness variations are smaller than those of the cepheids, usually less than a magnitude. Their periods are also shorter, less than a day. Like the W Vir stars, the RR Lyrae stars are old population II stars. They are very common in the globular star clusters and were therefore previously called cluster variables.

The absolute magnitudes of the RR Lyrae stars are about $M_V = 0.6 \pm 0.3$. They are all of roughly the same age and mass, and thus represent the same evolutionary phase, where helium is just beginning to burn in the core. Since the absolute magnitudes of the RR Lyrae variables are known, they can be used to determine distances to the globular clusters.

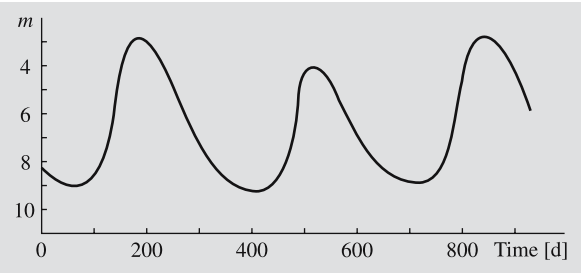

Fig. 13.5. The lightcurve of a long period Mira variable

Mira Variables (Fig. 13.5). The Mira variables (named after Mira Ceti) are supergiants of spectral classes M, S or C, usually with emission lines in their spectrum. They are losing gas in a steady stellar wind. Their periods are normally 100–500 days, and for this reason, they are also sometimes called long period variables. The amplitude of the light variations is typically about 6 magnitudes in the visual region. The period of Mira itself is about 330 days and its diameter is about 2 AU. At its brightest, Mira has the magnitude 2–4, but at light minimum, it may be down to 12. The effective temperature of the Mira variables is only about 2000 K. Thus 95% of their radiation is in the infrared, which means that a very small change in temperature can cause a very large change in visual brightness.

Other Pulsating Variables. One additional large group of pulsating stars are the *semiregular* and *irregular variables*. They are supergiants, often very massive young stars with unsteady pulsations in their extended outer layers. If there is some periodicity in the pulsations, these variables are called semiregular; otherwise they are irregular. An example of a semiregular variable is Betelgeuse (α Orionis). The pulsation mechanism of these stars is not well understood, since their outer layers are convective, and the theory of stellar convection is still poorly developed.

In addition to the main types of pulsating variables, there are some smaller separate classes, shown in Fig. 13.2.

The *dwarf cepheid* and the δ Scuti stars, which are sometimes counted as a separate type, are located below the RR Lyrae stars in the cepheid instability strip in the HR diagram. The dwarf cepheids are fainter and more rapidly varying than the classical cepheids. Their light curves often show a beating due to interference between the fundamental frequency and the first overtone.

The β *Cephei* stars are located in a different part of the HR diagram than the other variables. They are hot massive stars, radiating mainly in the ultraviolet. The variations are rapid and of small amplitude. The pulsation mechanism of the β Cephei stars is unknown.

The *RV Tauri* stars lie between the cepheids and the Mira variables in the HR diagram. Their period depends slightly on the luminosity. There are some unexplained features in the light curves of the RV Tauri stars, e. g. the minima are alternately deep and shallow.

13.3 Eruptive Variables

In the eruptive variables there are no regular pulsations. Instead sudden outbursts occur in which material is ejected into space. Nowadays such stars are divided into two main categories, *eruptive* and *cataclysmic variables*. Brightness changes of eruptive variables are caused by sudden eruptions in the chromosphere or corona, the contributions of which are, however, rather small in the stellar scale. These stars are usually surrounded by a gas shell or interstellar matter participating in the eruption. This group includes e. g. *flare stars*, various kinds of *nebular variables*, and *R Coronae Borealis stars*. Eruptions of the cataclysmic variables are due to nuclear reactions on the stellar surface or interior. Explosions are so violent that they can even destroy the whole star. This group includes *novae* and nova-like stars, *dwarf novae* and *supernovae* (Table 13.2).

Flare Stars. The *flare* or *UV Ceti stars* are dwarf stars of spectral class M. They are young stars, mostly found in young star clusters and associations. At irregular intervals there are flare outbursts on the surface of the stars similar to those on the Sun. The flares are related to disturbances in the surface magnetic fields. The energy of the outbursts of the flare stars is apparently about the same as in solar flares, but because the stars are much fainter than the Sun, a flare can cause a brightening by up to 4–5 magnitudes. A flare lights up in a few seconds and then fades away in a few minutes (Fig. 13.6). The same star may flare several times in one day. The optical flare is accompanied by a radio outburst, like in

Table 13.2. Main properties of eruptive variables (N, number of stars of the given type in Kukarkin's catalogue, Δm, change in brightness in magnitudes. The velocity is the expansion velocity in km/s, based on the Doppler shifts of the spectral lines)

Variable	N	Δm	Velocity
Supernovae	7	$\gtrsim 20$	4000–10000
Ordinary novae	210	7–18	200–3500
Recurrent novae		$\lesssim 10$	600
Nova-like stars (P Cygni, symbiotic)	80	$\lesssim 2$	30–100
Dwarf novae (SS Cyg = U Gem, ZZ Cam)	330	2–6	(700)
R Coronae Borealis	40	1–9	–
Irregular (nebular variables, T Tau, RW Aur)	1450	$\lesssim 4$	(300)
Flare stars (UV Ceti)	750	$\lesssim 6$	2000

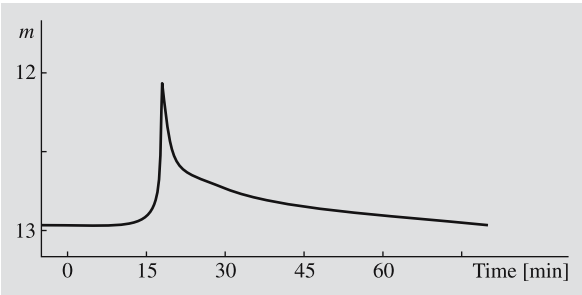

Fig. 13.6. The outbursts of typical flare stars are of short duration

the Sun. In fact, the flare stars were the first stars to be detected as radio sources.

Nebular Variables. In connection with bright and dark interstellar clouds e. g. in the constellations of Orion, Taurus and Auriga, there are variable stars. The *T Tauri stars* are the most interesting of them. These stars are newly formed or just contracting towards the main sequence. The brightness variations of the T Tauri stars are irregular (Fig. 13.7). Their spectra contain bright emission lines, formed in the stellar chromosphere, and forbidden lines, which can only be formed at extremely low densities. The spectral lines also show that matter is streaming out from the stars.

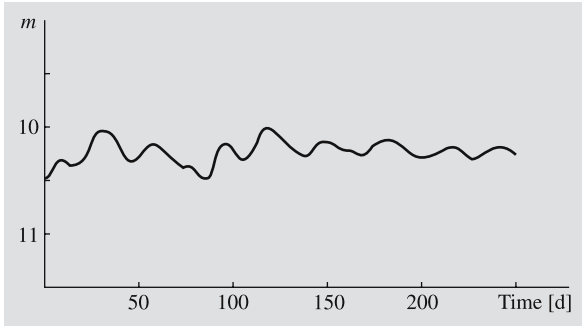

Fig. 13.7. Light curve of a T Tauri variable

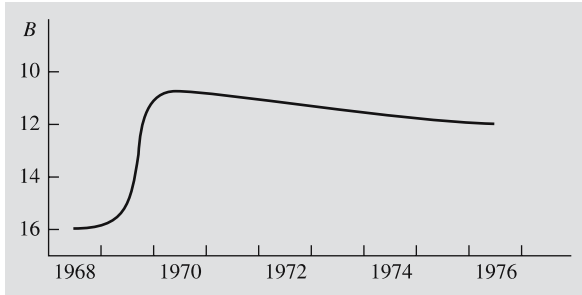

Fig. 13.8. In 1969–1970, the star V1057 Cygni brightened by almost 6 magnitudes

Since the T Tauri stars are situated inside dense gas clouds, they are difficult to observe. However, this situation has improved with the development of radio and infrared techniques.

Stars in the process of formation may change in brightness very rapidly. For example, in 1937, FU Orionis brightened by 6 magnitudes. This star is a strong source of infrared radiation, which shows that it is still enveloped by large quantities of interstellar dust and gas. A similar brightening by six magnitudes was observed in 1969 in V1057 Cygni (Fig. 13.8). Before its brightening, it was an irregular T Tauri variable; since then, it has remained a fairly constant tenth-magnitude AB star.

Stars of the *R Coronae Borealis* type have "inverse nova" light curves. Their brightness may drop by almost ten magnitudes and stay low for years, before the star brightens to its normal luminosity. For example, R CrB itself is of magnitude 5.8, but may fade to 14.8 magnitudes. Figure 13.9 shows its recent decline, based on observations by Finnish and French amateurs. The R CrB stars are rich in carbon and the decline is pro-

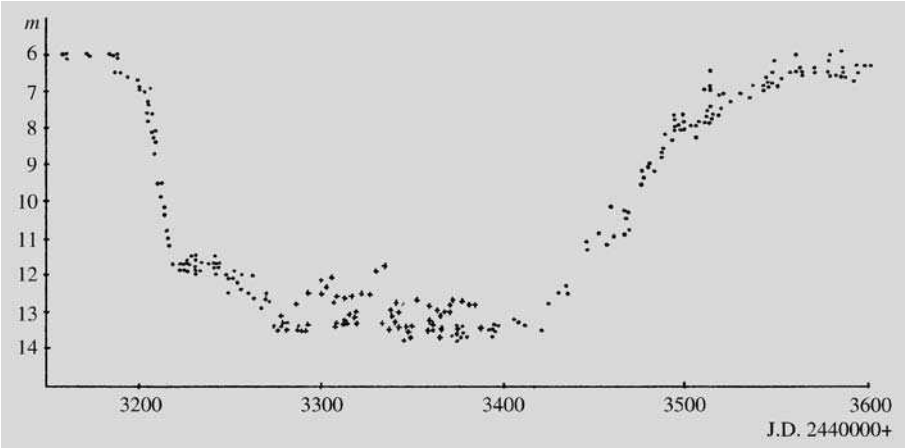

Fig. 13.9. The decline of R Coronae Borealis in 1977–1978; observations by Finnish and French amateur astronomers. (Kellomäki, Tähdet ja Avaruus 5/1978)

duced when the carbon condenses into a circumstellar dust shell.

One very interesting variable is η *Carinae* (Fig. 13.10). At present it is a six magnitude star surrounded by a thick, extensive envelope of dust and gas. In the early 19th century η Carinae was the second brightest star in the sky after Sirius. Around the middle of the century it rapidly dimmed to magnitude 8, but during the 20th century it has brightened somewhat. The circumstellar dust cloud is the brightest infrared source in the sky outside the solar system. The energy radiated by η Carinae is absorbed by the nebula and re-radiated at infrared wavelengths. It is not known whether η Carinae is related to the novae or whether it is a very young star that cannot evolve in the normal way because of the very thick cloud surrounding it.

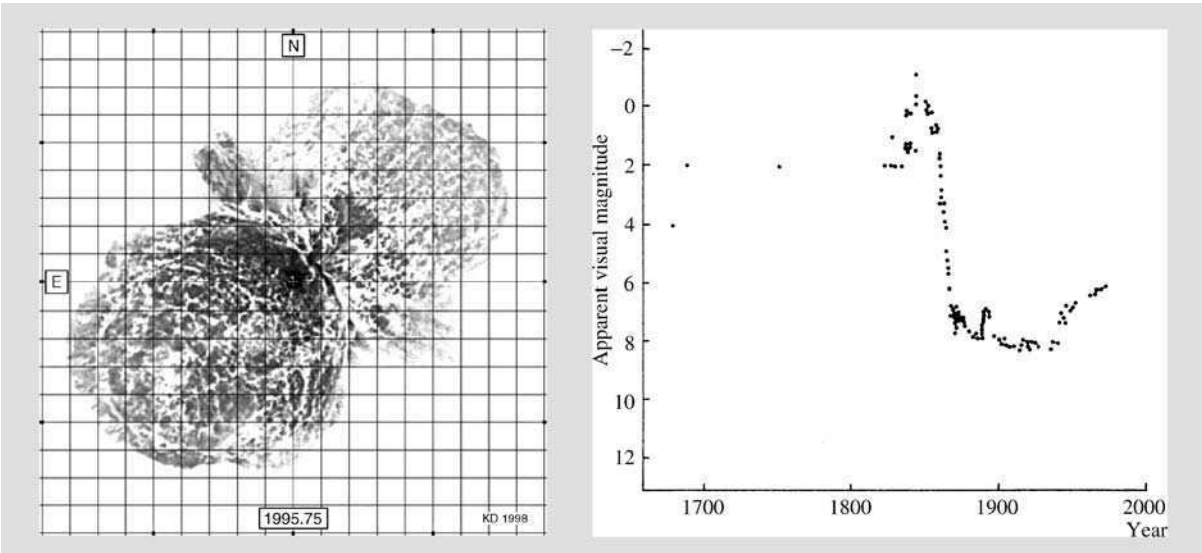

Fig. 13.10. In the 19th century, η Carinae was one of the brightest stars in the sky; since then it has dimmed considerably. In an outburst in 1843 the star ejected an expanding nebula, which has been called "Homunculus". The Hubble photograph has been printed negative, to show finer details in the bipolar outflow. (Photograph NASA/HST/University of Minnesota)

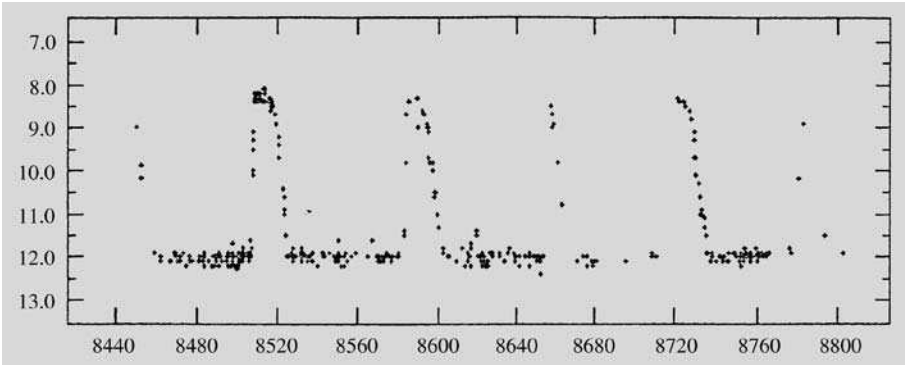

Fig. 13.11. The light curve of the dwarf nova SS Cygni in the beginning of 1966. (Drawing by Martti Perälä is based on observations by Nordic amateurs)

Novae. One of the best known types of eruptive variables are the *novae*. They are classified into several subtypes: *ordinary novae*, *recurrent novae* and *nova-like variables*. The *dwarf novae* (Fig. 13.11) are nova-like rather frequently eruptive stars with some special characteristics. They are nowadays divided into three subgroups.

The outbursts of all novae are rapid. Within a day or two the brightness rises to a maximum, which may be 7–18 magnitudes brighter than the normal luminosity. This is followed by a gradual decline, which may go on for months or years. The light curve of a typical nova is shown in Fig. 13.12. This light curve of Nova Cygni 1975 has been composed from hundreds of observations, mostly by amateurs.

In recurrent novae, the brightening is somewhat less than 10 magnitudes and in dwarf novae, 2–6 magnitudes. In both types there are repeated outbursts. For recurrent novae the time between outbursts is a few decades and for the dwarf novae 20–600 days. The interval depends on the strength of the outburst: the stronger the outburst, the longer the time until the next one. The brightening in magnitudes is roughly proportional to the logarithm of the recharging interval. It is possible that ordinary novae obey the same relationship. However, their amplitude is so much larger that the time between outbursts should be thousands or millions of years.

Observations have shown all novae and dwarf novae to be members of close binary systems. One component of the system is a normal star and the other is a white dwarf surrounded by a gas ring. (The evolution of close binary systems was considered in Sect. 11.6, where it was seen how this kind of system might have been formed.) The normal star fills its Roche surface, and material from it streams over to the white dwarf. When enough mass has collected on the surface of the white dwarf, the hydrogen is explosively ignited and the outer shell is ejected. The brightness of the star grows rapidly. As the ejected shell expands, the temperature of the star drops and the luminosity gradually decreases. However, the outburst does not stop the mass transfer from the companion star, and gradually the white dwarf accretes new material for the next explosion (Fig. 13.13).

The emission and absorption lines from the expanding gas shell can be observed in the spectrum of a nova. The Doppler shifts correspond to an expansion velocity of about 1000 km/s. As the gas shell disperses, its spectrum becomes that of a typical diffuse emission nebula. The expanding shell around a nova can also sometimes be directly seen in photographs.

A considerable fraction of the novae in our Galaxy are hidden by interstellar clouds and their number is therefore difficult to estimate. In the Andromeda Galaxy, observations indicate 25–30 nova explosions per year. The number of dwarf novae is much larger. In addition there are nova-like variables, which share many of the properties of novae, such as emission lines from circumstellar gas and rapid brightness variations. These variables, some of which are called *symbiotic stars*, are close binaries with mass transfer. Gas streaming from the primary hits a gas disc around the secondary in a hot spot, but there are no nova outbursts.

Supernovae. The supernovae are stars with the largest brightness variations. In a few days their brightness may rise more than 20 magnitudes, i. e. the luminosity may increase by a factor of a hundred million. After the maximum, there is a slow decline lasting several years.

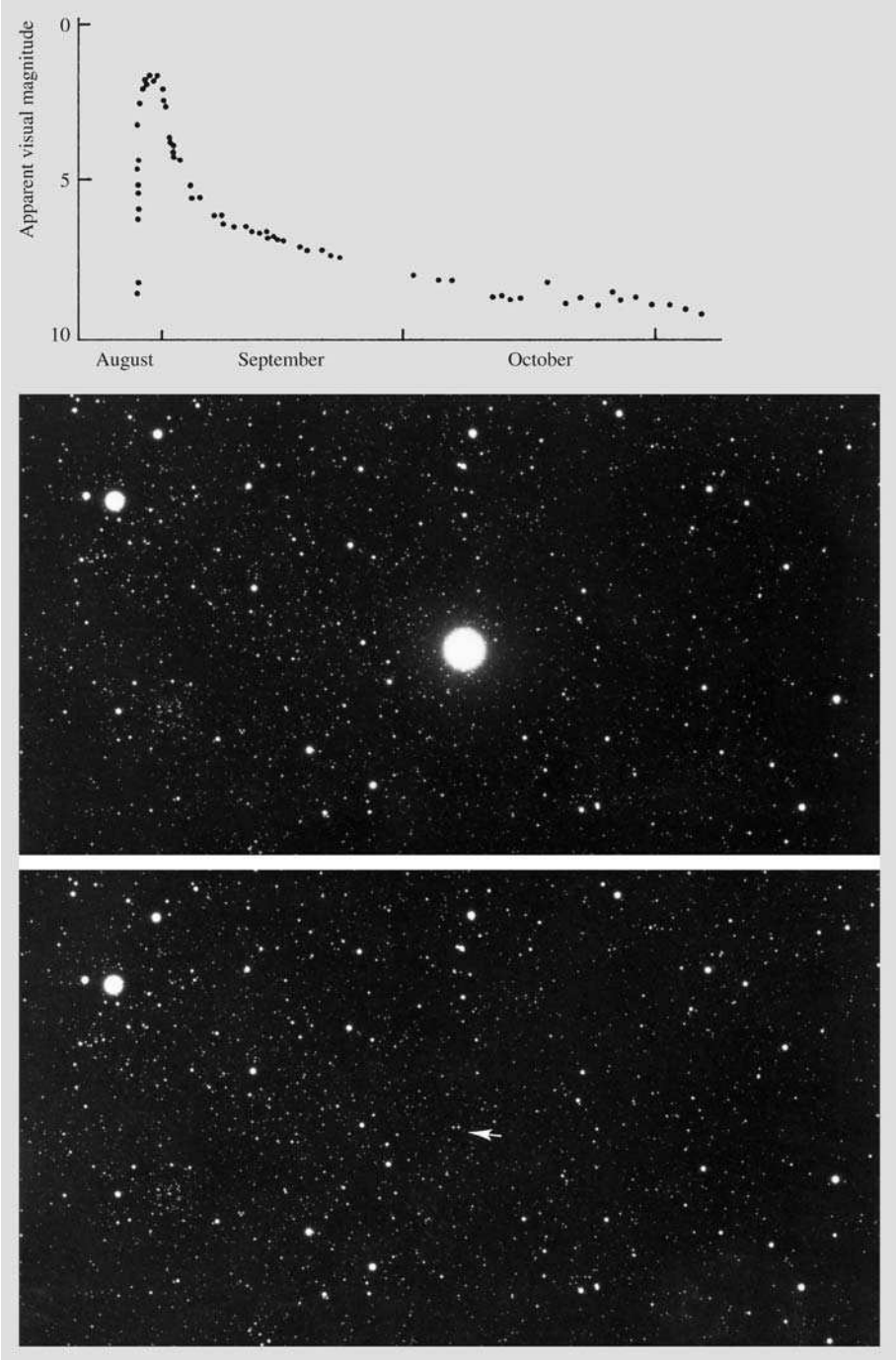

Fig. 13.12. In 1975 a new variable, Nova Cygni or V1500 Cygni, was discovered in Cygnus. In the upper photograph the nova is at its brightest (about 2 magnitudes), and in the lower photograph it has faded to magnitude 15. (Photographs Lick Observatory)

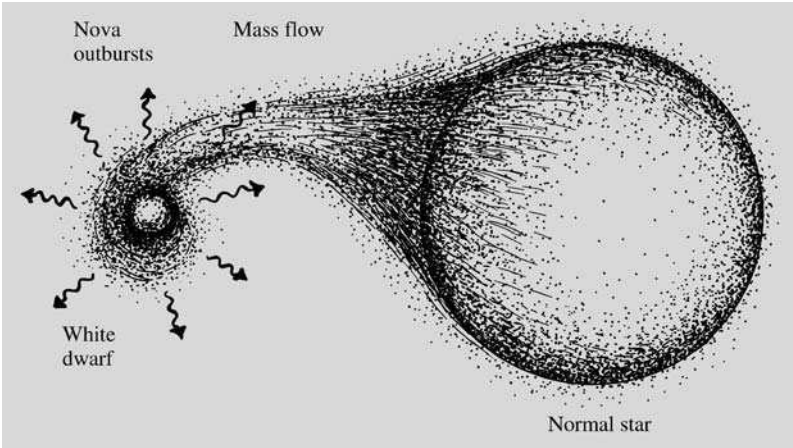

Fig. 13.13. The novae are thought to be white dwarfs accreting matter from a nearby companion star. At times, nuclear reactions burning the accreted hydrogen are ignited, and this is seen as the flare-up of a nova

The supernovae are exploding stars. In the explosion a gas shell expanding with a velocity around 10,000 km/s is ejected. The expanding gas shell remains visible for thousands of years. A few tens of such supernova remnants have been discovered in the Milky Way. The remnant of the actual star may be a neutron star or a black hole.

The supernovae are classified as type I and type II based on whether their spectra show evidence of hydrogen. Type I supernovae are further divided into types Ia (silicon is present), Ib (no silicon, but helium present), and Ic (no silicon, little helium).

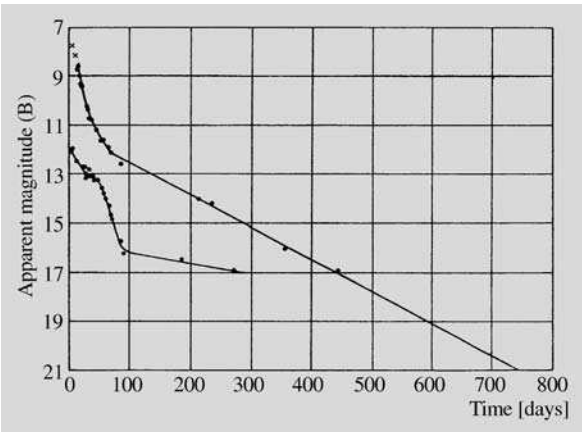

Fig. 13.14. The lightcurves of the two types of supernovae. Above SN 1972e type I and below SN 1970g type II. (Kirchner, R.P. (1976): Sci. Am. **235**, No. 6, 88)

The supernova types also differ in regard to their light curves (Fig. 13.14). *Type I* supernovae fade away in a regular manner, almost exponentially. The decline of a *type II* supernova is less regular and its maximum luminosity is smaller. Type II supernovae have been further subdivided into several classes on the basis of their light curves.

In Chap. 11 it was mentioned that there are several possible ways in which a star may come to explode at the end of its evolution. The type II supernova is the natural endpoint of the evolution of a single star. Types Ib and Ic also arise from massive stars that have lost their hydrogen envelope by winds or to a companion (e. g. Wolf–Rayet stars). These types are collectively referred to as *core-collapse supernovae*.

The type Ia supernovae, on the other hand, have about the mass of the Sun and should end up as white dwarfs. However, if a star is accreting mass from a binary companion, it will undergo repeated nova outbursts. Some of the accreted material will then be turned into helium or carbon and oxygen, and will collect on the star and increase its mass. Finally the mass may exceed the Chandrasekhar limit. The star will then collapse and explode as a supernova.

At least six supernova explosions have been observed in the Milky Way. Best known are the "guest star" seen in China in 1054 (whose remnant is the Crab nebula), Tycho Brahe's supernova in 1572 and Kepler's supernova in 1604. On the basis of observations of other Sb–Sc-type spiral galaxies, the interval between supernova explosions in the Milky Way is

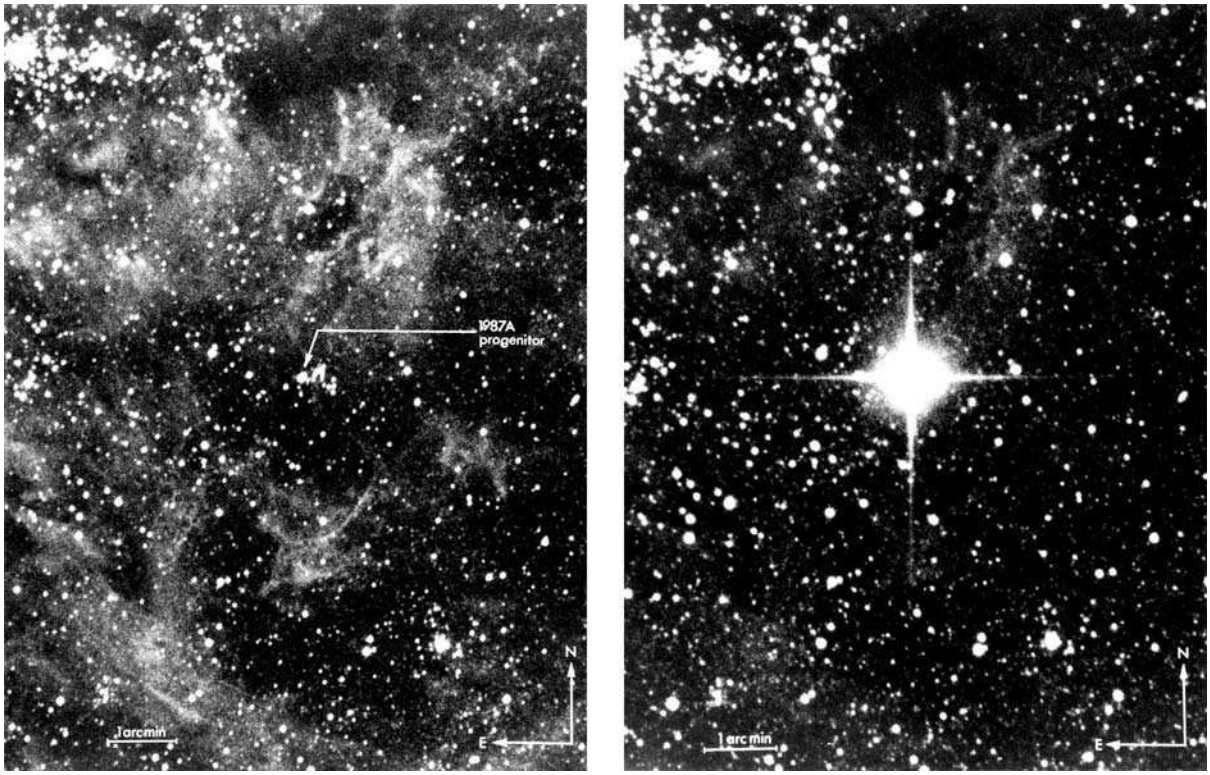

Fig. 13.15. Supernova 1987A in the Large Magellanic Cloud before and after the explosion. (Photographs ESO)

predicted to be about 50 years. Some will be hidden by obscuring material, but the 400 years' interval since the last observed supernova explosion is unusually long.

On February 23, 1987 the first burst of light from a supernova in the Large Magellanic Cloud, the small companion galaxy of the Milky Way, reached the Earth (Fig. 13.15). This supernova, SN 1987A, was of type II, and was the brightest supernova for 383 years. After its first detection, SN 1987A was studied in great detail by all available means. Although the general ideas of Sects. 11.4 and 11.5 on the final stages of stellar evolution have been confirmed, there are complications. Thus e. g. the progenitor star was a blue rather than a red giant as expected, perhaps because of the lower abundance of heavy elements in the Large Magellanic Cloud compared to that in the Milky Way. The collapse of its core released a vast amount of energy as a pulse of neutrinos, which was detected in Japan and the USA. The amount of energy released indicates that the remnant is a neutron star. As of January 2006 the remnant has, however, not been seen yet.

13.4 Examples

Example 13.1 The observed period of a cepheid is 20 days and its mean apparent magnitude $m = 20$. From Fig. 13.4, its absolute magnitude is $M \approx -5$. According to (4.11), the distance of the cepheid is

$$r = 10 \times 10^{(m-M)/5} = 10 \times 10^{(20+5)/5}$$
$$= 10^6 \text{ pc} = 1 \text{ Mpc} .$$

Example 13.2 The brightness of a cepheid varies 2 mag. If the effective temperature is 6000 K at the maximum and 5000 K at the minimum, how much does the radius change?

The luminosity varies between

$$L_{\max} = 4\pi R_{\max}^2 \sigma T_{\max}^4 ,$$

$$L_{\min} = 4\pi R_{\min}^2 \sigma T_{\min}^4 .$$

In magnitudes the difference is

$$\Delta m = -2.5 \lg \frac{L_{\min}}{L_{\max}} = -2.5 \lg \frac{4\pi R_{\min}^2 \sigma T_{\min}^4}{4\pi R_{\max}^2 \sigma T_{\max}^4}$$

$$= -5 \lg \frac{R_{\min}}{R_{\max}} - 10 \lg \frac{T_{\min}}{T_{\max}} .$$

This gives

$$\lg \frac{R_{\min}}{R_{\max}} = -0.2\Delta m - 2 \lg \frac{T_{\min}}{T_{\max}}$$

$$= -0.4 - 2 \lg \frac{5000}{6000} = -0.24 ,$$

whence

$$\frac{R_{\min}}{R_{\max}} = 0.57 .$$

13.5 Exercises

Exercise 13.1 The absolute visual magnitude of RR Lyrae variables is 0.6 ± 0.3. What is the relative error of distances due to the deviation in the magnitude?

Exercise 13.2 The bolometric magnitude of a long period variable varies by one magnitude. The effective temperature at the maximum is 4500 K.

a) What is the temperature at the minimum, if the variation is due to temperature change only?

b) If the temperature remains constant, what is the relative variation in the radius?

Exercise 13.3 In 1983 the radius of the Crab nebula was about $3'$. It is expanding $0.21''$ a year. Radial velocities of $1300\,\mathrm{km\,s^{-1}}$ with respect to the central star have been observed in the nebula.

a) What is the distance of the nebula, assuming its expansion is symmetric?

b) A supernova explosion has been observed in the direction of the nebula. Estimate, how long time ago?

c) What was the apparent magnitude of the supernova, if the absolute magnitude was a typical -18?